Elusive 'usual' warm-up in injury prevention trials: the problem of comparing apples to beans

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THE ELUSIVE 'USUAL' WARM-UP

Comparing apples to oranges is rational when comparing different fruits; comparing apples to beans is something else entirely. Likewise, injury prevention trials proposing warm-up interventions should strive to deliver reasonable comparisons. The contents and dosage of the intervention and control warm-ups may vary, especially in the context of pragmatic trials, but should be appropriately described to afford transparency and comparability. So, what is this elusive entity known as the 'usual' warm-up in injury prevention trials? And how may this affect the interpretation of the findings? This editorial explores the context and importance of the 'usual' warm-up in injury prevention trials and provides recommendations to harmonise future research.

WHAT EXACTLY IS BEING COMPARED?

Warm-up protocols focused on injury prevention are often compared with other warm-ups to assess their relative effectiveness. Pragmatic trials often compare a standardised protocol (the intervention) to 'usual' warm-up (the comparator), and this is relevant to reflect current real-world practices. However, accurate evaluation of effectiveness requires clear descriptions of the content of the comparator warm-up protocols. Inadequate and incomplete reporting is more common than desirable, even in recent randomised studies published in high-profile journals.[1–3] We describe three examples in detail to illustrate this problem.

Example 1: A cluster randomised trial compared the effectiveness of the warm-up programme 'VolleyVeilig' in reducing injuries (35 teams, n=282 players) to 'usual' warm-up (31 teams, n=236 players).[1] While the 'VolleyVeilig' programme was described in detail, there was no information regarding the 'usual' warm-up (eg, contents, dosage), except that it was the teams' regular warm-up routine.

Example 2: A cluster randomised trial assessed the effects of the FIFA '11+Kids' warm-up (128 teams, n=2066 players) on injury prevention compared with 'usual' warm-up (115 teams, n=1829 players).[2] The '11+Kids' was described in detail, but no information was provided about the 'usual' warm-up. Since the '11+Kids' was implemented twice a week in *replacement* of 'usual' warm-up, it might be inferred that they had similar duration, but no other information can be retrieved. Moreover, the difficulty of the exercises used in the '11+Kids' was progressively increased during the intervention, but no information was available on whether progressions were implemented in the 'usual' warm-up as well.

Example 3: A study compared the 'SHRed Injuries Basketball' warm-up (season 2; 31 teams, n=307 players) to 'standard-of-practice' warm-up (season 1; 63 teams, n=502 players) in youth basketball players.[3] Again, no information was provided regarding the 'standard' warm-up (eg, contents, dosage).

These selected examples serve to illustrate a wider problem across the injury prevention research landscape, with similar challenges occurring in studies with adult athletes.[4 5] Additional examples are presented as online supplemental material.

WHY IS THIS RELEVANT?

The warm-up may acutely influence performance and injury risk,[6] but in the absence of specific descriptions the terms 'usual' or 'standard' or 'regular' warm-up are too generic and have no informative value, impair reproducibility and affect interstudy comparability. There are multiple strategies to organise a warm-up programme regarding its contents and dosage, and teams may wish to implement varying warm-ups in different weekdays and/or periods of the season.[7] Therefore, the 'usual' warm-up may diverge considerably from team to team and even within the same team based on daily goals and/or individualisation concerns.[4 7] This is to be expected, reflects real-life complexities and diversity and can be compared with 'usual care' in healthcare settings. When lumping and comparing distinct warm-ups, however, we risk pooling data that should not be pooled, generating a non-representative average value leading to erroneous interpretations. This underreporting and underdescription limits the interpretation of the findings and impairs comparability across studies.

MOVING FORWARD: IMPROVE THE REPORTING OF THE 'USUAL' WARM-UP

The warm-up contents may change its effects[6 8] and should be reported in detail, as should their sequence (ie, temporal order). When the contents are *exactly* the same, changing a single parameter such as warm-up duration may change its effects, as was recently demonstrated.[9] Therefore, proper reporting of the 'usual' warm-up should consider content-related and dosage-related information. An illustrative example of simple but informative reporting is provided in Ding *et al*'s[10] study, where the 'standard' warm-up consisted of running and stretching and was described as having the same duration as the experimental warm-up.

Progressions or changes to the 'usual' warm-up across the experimental period should also be reported to more accurately ascertain what is being compared. Ideally, whether the 'usual' warm-up is collective, individualised or a mixture of both should also be reported. Compliance, adherence and/or fidelity should be assessed and reported, as these may change the effects of a warm-up programme in injury prevention.[2] It is not enough to report this feature only for the experimental group.[2] Unbalanced compliance, adherence and/or fidelity between 'usual' and experimental warm-up is expected to affect the results

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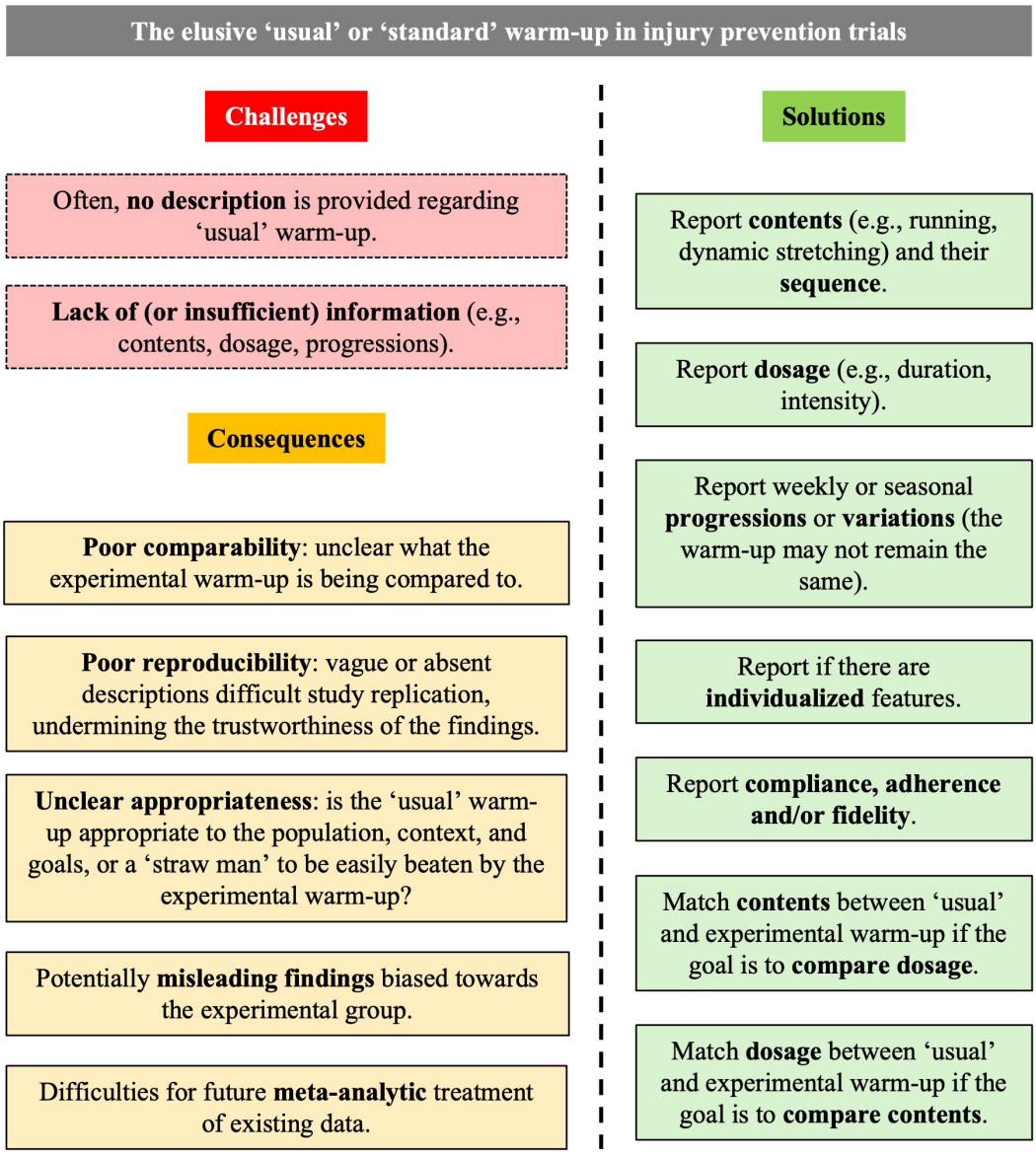

Figure 1 Challenges and proposed solutions to the under-reporting of 'usual' warm-up.

and should be reported for all groups, including the 'usual' warm-up group.

Proper reporting of the 'usual' warm-up is even more relevant in cluster randomised trials, as different teams may use very distinct warm-ups,[4 7] and this has a high risk of confounding the results (performance bias). We are in favour of trials allowing comparator groups to use their 'usual' warm-up, but in the absence of a minimal set of data concerning the 'usual' warm-up, we may be comparing apples to beans. Most journals allow the submission of electronic supplementary materials that provide an opportunity for a more complete reporting of the 'usual' warm-up. Notwithstanding, when different warm-ups are being compared, the 'usual' warm-up becomes a core aspect of the methodology and should be described within the confines of the main manuscript. Figure 1 synthesises the challenges and our proposed solutions.

COMPLETENESS OF REPORTING MATTERS

As the number of scientific publications increases at startling rates, it is timely to join efforts to promote a higher quality of research reporting while respecting the pragmatic setting of trials. Completeness of reporting of core elements of a study design will afford better informed interpretation of the data. The elusive 'usual' warm-up should be accompanied by an appropriate description of its features to provide a more accurate interpretation of the findings and facilitate comparability among different studies and in meta-analyses.

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